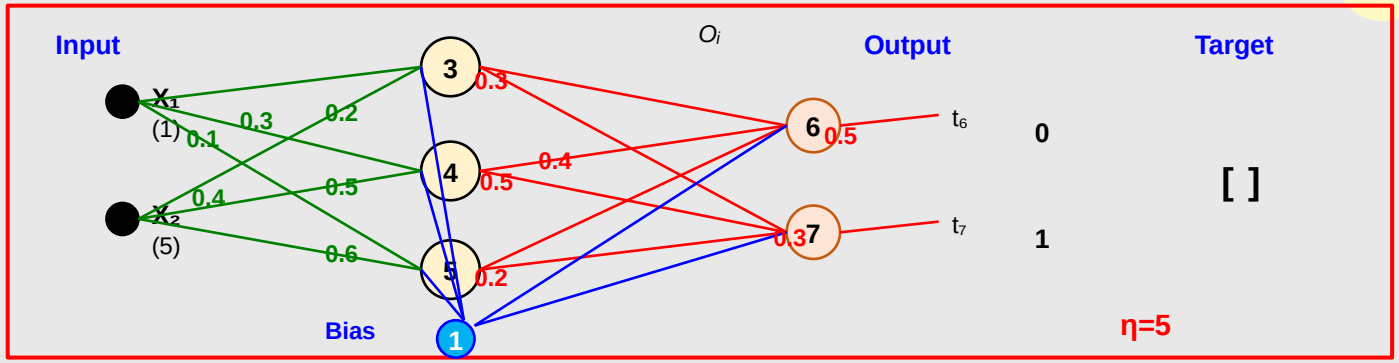
$$\delta_j = O_j(1 - O_j)(t_j - O_j)$$

$$\Delta_{ji} = \eta \cdot O_i \cdot O_j(1 - O_j)(t_j - O_j) = \eta \cdot O_i \cdot \delta_j$$



Backward (Update Output Layer Weights)

$\delta_6 = 0.791(1 - 0.791)(0 - 0.791) = -0.131$

$\delta_7 = 0.791(1 - 0.791)(1 - 0.791) = 0.035$

$\Delta W_{63} = 5 * (-0.131) * 0.9 = -0.59$

$\Delta W_{64} = 5 * (-0.131) * 0.943 = -0.618$

$\Delta W_{65} = 5 * (-0.131) * 0.968 = -0.634$

$\Delta W_{6}b = 5 * (-0.131) * 1.0 = -0.655$

$W_{63} = 0.3 - 0.59 = -0.29$

$W_{64} = 0.4 - 0.618 = -0.218$

$W_{65} = 0.5 - 0.634 = -0.134$

$W_{6}b = 0.2 - 0.655 = -0.455$

$\Delta W_{73} = 5 * (0.035) * 0.9 = 0.158$

$\Delta W_{74} = 5 * (0.035) * 0.943 = 0.165$

$\Delta W_{75} = 5 * (0.035) * 0.968 = 0.169$

$\Delta W_{7}b = 5 * (0.035) * 1.0 = 0.175$

$W_{73} = 0.3 + 0.158 = 0.458$

$W_{74} = 0.4 + 0.165 = 0.565$

$W_{75} = 0.5 + 0.169 = 0.669$

$W_{7}b = 0.2 + 0.175 = 0.375$

$$O_3 = \frac{1}{1 + e^{-2.2}} = 0.9$$

$$O_4 = \frac{1}{1 + e^{-2.8}} = 0.943$$

$$O_5 = \frac{1}{1 + e^{-3.4}} = 0.968$$

$$O_6 = \frac{1}{1 + e^{-1.3312}} = 0.791$$

$$O_7 = \frac{1}{1 + e^{-1.3312}} = 0.791$$